

Unitree Week 21

Presentation by Mario Ortiz

This Week

- Unitree R1 Implementation to IsaacLab Reinforcement Learning rough_env.cfg code.
 -

R1 Reinforcement Learning Program Implementation

1. Need to obtain Unitree R1 CFG contents

```
##  
from isaacsim_assets import G1_MINIMAL_CFG
```

Categories of Data

- R1 EDU USD Path
- R1 EDU Joint names
- R1 EDU body/link names
- R1 EDU joint limits
- R1 Edu actuator gains

Convert URDF File to UDF file

- Unitree R1 URDF file → located in the **unitree_ros** repository
 - As Unitree R1 USD file is not publicized, we will convert the R1 URDF into the USD file using the Isaac Sim converter.
 - 1. Run Isaac Sim (`.\isaacclab_bat -s`)
 - 2. File → Import
 - 3. Choose the R1 URDF file
 - 4. Choose the file you want to put your converted UDF file to
 - 5. Set up parameters:

```
PS C:\Users\Mario\IsaacLab\source\isaacsim_tasks\isaacsim_tasks\manager_based\locomotion\velocity\config\r1\assets\R1> Get-ChildItem -Recurse
```

Directory:

C:\Users\Mario\IsaacLab\source\isaacsim_tasks\isaacsim_tasks\manager_based\locomotion\velocity\config\r1\assets\R1

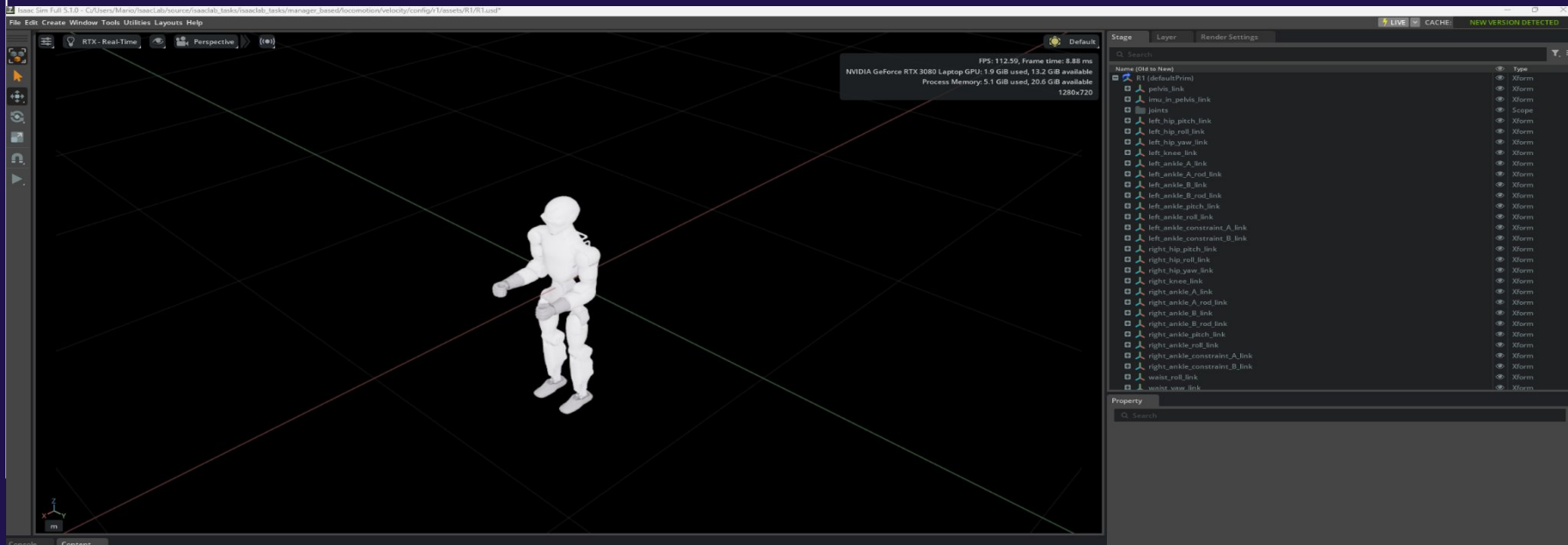
Mode	LastWriteTime	Length	Name
d----	6/25/2026 8:13 PM		configuration
-a----	6/25/2026 8:13 PM	1448	R1.usd

Directory: C:\Users\Mario\IsaacLab\source\isaacsim_tasks\isaacsim_tasks\manager_based\locomotion\velocity\config\r1\assets\R1\configuration

Mode	LastWriteTime	Length	Name
-a----	6/25/2026 8:13 PM	32373007	R1_base.usd
-a----	6/25/2026 8:13 PM	13198	R1_physics.usd
-a----	6/25/2026 8:13 PM	3013	R1_robot.usd
-a----	6/25/2026 8:13 PM	642	R1_sensor.usd

Opening USD file on IsaacSim

- Launch IsaacSim, then open the file in which the R1 usd file is located in:



Opening USD file on IsaacSim

- Testing initial asset file (like the G1_Minimal_CFG asset file)



R1 Asset File

- Now that we have the Unitree's asset file, which will be used to load the usd file onto IsaacSim for training.

Opening USD file on IsaacSim

```
=====
R1 LOADED SUCCESSFULLY
=====
```

```
Number of joints: 26
```

```
JOINT NAMES
-----
```

```
00: left_hip_pitch_joint
01: right_hip_pitch_joint
02: waist_roll_joint
03: left_hip_roll_joint
04: right_hip_roll_joint
05: waist_yaw_joint
06: left_hip_yaw_joint
07: right_hip_yaw_joint
08: head_pitch_joint
09: left_shoulder_pitch_joint
10: right_shoulder_pitch_joint
11: left_knee_joint
12: right_knee_joint
13: head_yaw_joint
14: left_shoulder_roll_joint
15: right_shoulder_roll_joint
16: left_ankle_pitch_joint
17: right_ankle_pitch_joint
18: left_shoulder_yaw_joint
19: right_shoulder_yaw_joint
20: left_ankle_roll_joint
21: right_ankle_roll_joint
22: left_elbow_joint
23: right_elbow_joint
24: left_wrist_roll_joint
25: right_wrist_roll_joint
```

```
Number of bodies: 40
```

```
BODY NAMES
-----
```

```
00: pelvis_link
01: imu_in_pelvis_link
02: left_hip_pitch_link
03: right_hip_pitch_link
04: waist_roll_link
05: left_hip_roll_link
06: right_hip_roll_link
07: waist_yaw_link
08: left_hip_yaw_link
09: right_hip_yaw_link
10: head_pitch_link
11: left_shoulder_pitch_link
12: right_shoulder_pitch_link
13: left_knee_link
14: right_knee_link
15: head_yaw_link
16: left_shoulder_roll_link
17: right_shoulder_roll_link
18: left_ankle_A_link
19: left_ankle_B_link
20: left_ankle_pitch_link
21: right_ankle_A_link
22: right_ankle_B_link
23: right_ankle_pitch_link
24: left_shoulder_yaw_link
25: right_shoulder_yaw_link
26: left_ankle_A_rod_link
27: left_ankle_B_rod_link
28: left_ankle_roll_link
29: right_ankle_A_rod_link
30: right_ankle_B_rod_link
31: right_ankle_roll_link
32: left_elbow_link
33: right_elbow_link
34: left_ankle_constraint_A_link
35: left_ankle_constraint_B_link
36: right_ankle_constraint_A_link
37: right_ankle_constraint_B_link
38: left_wrist_roll_link
39: right_wrist_roll_link
```

```
DEFAULT JOINT POSITIONS
-----
```

```
left_hip_pitch_joint      : 0.000000
right_hip_pitch_joint     : 0.000000
waist_roll_joint          : 0.000000
left_hip_roll_joint       : 0.000000
right_hip_roll_joint      : 0.000000
waist_yaw_joint           : 0.000000
left_hip_yaw_joint        : 0.000000
right_hip_yaw_joint       : 0.000000
head_pitch_joint          : 0.000000
left_shoulder_pitch_joint : 0.000000
right_shoulder_pitch_joint : 0.000000
left_knee_joint           : 0.000000
right_knee_joint          : 0.000000
head_yaw_joint            : 0.000000
left_shoulder_roll_joint  : 0.000000
right_shoulder_roll_joint : 0.000000
left_ankle_pitch_joint    : 0.000000
right_ankle_pitch_joint   : 0.000000
left_shoulder_yaw_joint   : 0.000000
right_shoulder_yaw_joint  : 0.000000
left_ankle_roll_joint     : 0.000000
right_ankle_roll_joint    : 0.000000
left_elbow_joint          : 0.000000
right_elbow_joint         : 0.000000
left_wrist_roll_joint     : 0.000000
right_wrist_roll_joint    : 0.000000
```

Opening USD file on IsaacSim

- Information obtained: number of joints (26), body names (40 rigid bodies), default joint positions, and soft joint position limits.
- Different naming conventions from the G1 (example: G1's "torso_link" is referred as R1's "pelvis_link")

SOFT JOINT POSITION LIMITS

```
-----  
left_hip_pitch_joint      : lower=-2.658136, upper= 2.274164  
right_hip_pitch_joint     : lower=-2.658136, upper= 2.274164  
waist_roll_joint          : lower=-0.471240, upper= 0.471240  
left_hip_roll_joint       : lower=-0.907571, upper= 1.605703  
right_hip_roll_joint      : lower=-1.605703, upper= 0.907571  
waist_yaw_joint           : lower=-2.356200, upper= 2.356200  
left_hip_yaw_joint        : lower=-2.466180, upper= 2.466180  
right_hip_yaw_joint       : lower=-2.466180, upper= 2.466180  
head_pitch_joint          : lower=-0.565488, upper= 0.565488  
left_shoulder_pitch_joint : lower=-2.879800, upper= 1.832600  
right_shoulder_pitch_joint : lower=-2.879800, upper= 1.832600  
left_knee_joint           : lower=-0.044506, upper= 2.295981  
right_knee_joint          : lower=-0.044503, upper= 2.295981  
head_yaw_joint            : lower=-1.806390, upper= 1.806390  
left_shoulder_roll_joint  : lower=-0.091626, upper= 2.343135  
right_shoulder_roll_joint : lower=-2.343225, upper= 0.091535  
left_ankle_pitch_joint    : lower=-0.800229, upper= 0.503529  
right_ankle_pitch_joint   : lower=-0.800229, upper= 0.503529  
left_shoulder_yaw_joint   : lower=-1.727910, upper= 1.727910  
right_shoulder_yaw_joint  : lower=-1.727910, upper= 1.727910  
left_ankle_roll_joint     : lower=-0.235620, upper= 0.235620  
right_ankle_roll_joint    : lower=-0.235619, upper= 0.235619  
left_elbow_joint          : lower=-0.817598, upper= 2.027158  
right_elbow_joint         : lower=-0.817598, upper= 2.027158  
left_wrist_roll_joint     : lower=-1.727910, upper= 1.727910  
right_wrist_roll_joint    : lower=-1.727910, upper= 1.727910
```

Reinforcement Learning Testing

- How G1's rough_env_cfg is updated for the use of the R1:
 - Replacing input asset file (R1_EDU_CFG)
 - Switch G1 parameters to R1 parameters (or confirm already used G1 parameters can be used):
 - torso_link → pelvis_link
 - ankle_roll_link → compatible with R1
 - Utilizing flat environment for initial testing
 - For now, disabling height scanner (adding it back once we work on rough terrains)
 - Lower command ranges such as velocity
 - Disabled randomization commands

Reinforcement Learning Testing (continued)

- Training policy: since the G1's `rsl_rl_ppo.py` code consists only training parameters, we can use the same code to train the Unitree R1 for the time being.

Important Debugging Notes

- Specific Package Versions:

```
PS C:\Users\Mario\IsaacLab> .\isaacclab.bat -p .\scripts\r1\check_r1_package_versions.py  
[INFO] Using python from: C:\Users\Mario\IsaacLab\_\isaac_sim\python.bat
```

```
RL PACKAGE VERSIONS
```

```
=====
```

torch	: 2.7.0+cu128
tensorDict	: 0.9.0
torchrl	: NOT FOUND
rs1-rl-lib	: 5.0.1
h5py	: 3.11.0
numpy	: 1.26.4
isaacclab	: 0.54.3
isaacclab_r1	: 0.5.0

Testing RL pipeline (confirm it runs)

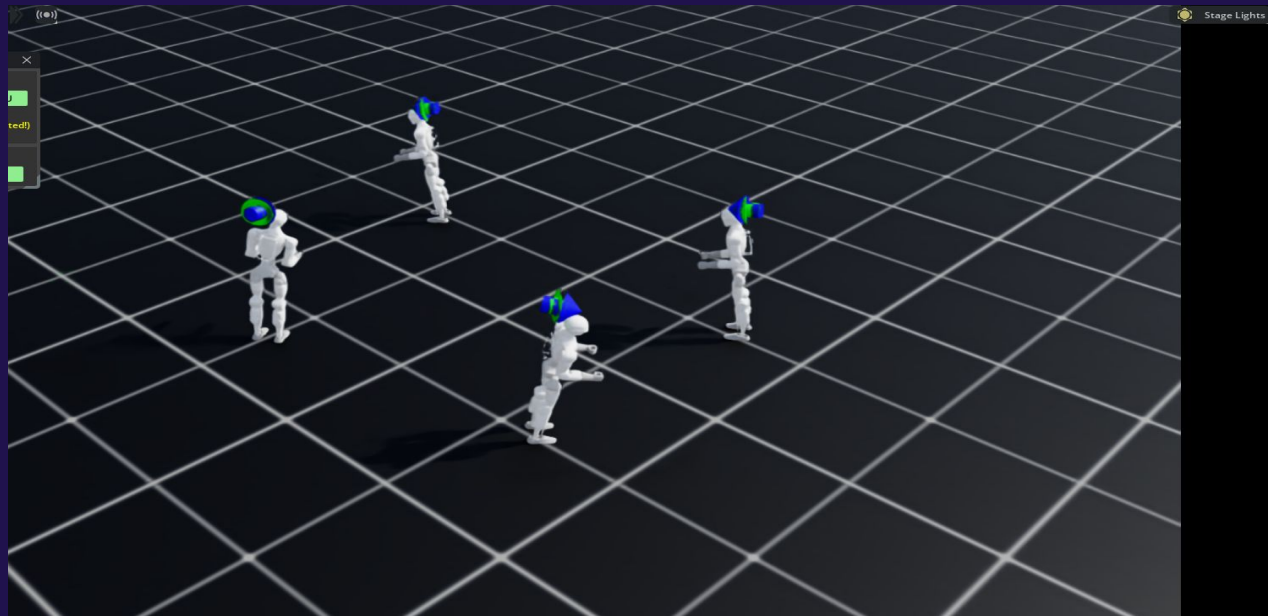
- Command:
.\isaacclab.bat -p
.\scripts\reinforcement_learning\rsl_rl\train.py --task
Isaac-Velocity-Rough-R1-v0 --num_envs
16 --max_iterations 1
--headless

```
/
Storing git diff for 'IsaacLab' in: C:\Users\Mario\IsaacLab\logs\rsl_rl\r1_rough\2026-06-26_09-09-52\git\IsaacLab.dif
f
#####
                        Learning iteration 0/1
#####
                        Total steps: 384
                        Steps per second: 196
                        Collection time: 1.653s
                        Learning time: 0.306s
                        Mean value loss: 0.1991
                        Mean surrogate loss: 0.0436
                        Mean entropy loss: 36.8920
                        Mean action std: 1.00
Episode_Reward/track_lin_vel_xy_exp: 0.0000
Episode_Reward/track_ang_vel_z_exp: 0.0000
Episode_Reward/ang_vel_xy_l2: 0.0000
Episode_Reward/dof_torques_l2: 0.0000
Episode_Reward/dof_acc_l2: 0.0000
Episode_Reward/action_rate_l2: 0.0000
Episode_Reward/feet_air_time: 0.0000
Episode_Reward/flat_orientation_l2: 0.0000
Episode_Reward/dof_pos_limits: 0.0000
Episode_Reward/termination_penalty: 0.0000
Episode_Reward/feet_slide: 0.0000
Episode_Reward/joint_deviation_hip: 0.0000
Episode_Reward/joint_deviation_waist: 0.0000
Episode_Reward/joint_deviation_arms: 0.0000
Episode_Reward/joint_deviation_head: 0.0000
Metrics/base_velocity/error_vel_xy: 0.0000
Metrics/base_velocity/error_vel_yaw: 0.0000
Episode_Termination/time_out: 0.0000
Episode_Termination/base_contact: 0.0000
-----
                        Iteration time: 1.96s
                        Time elapsed: 00:00:01
                        ETA: 00:00:00

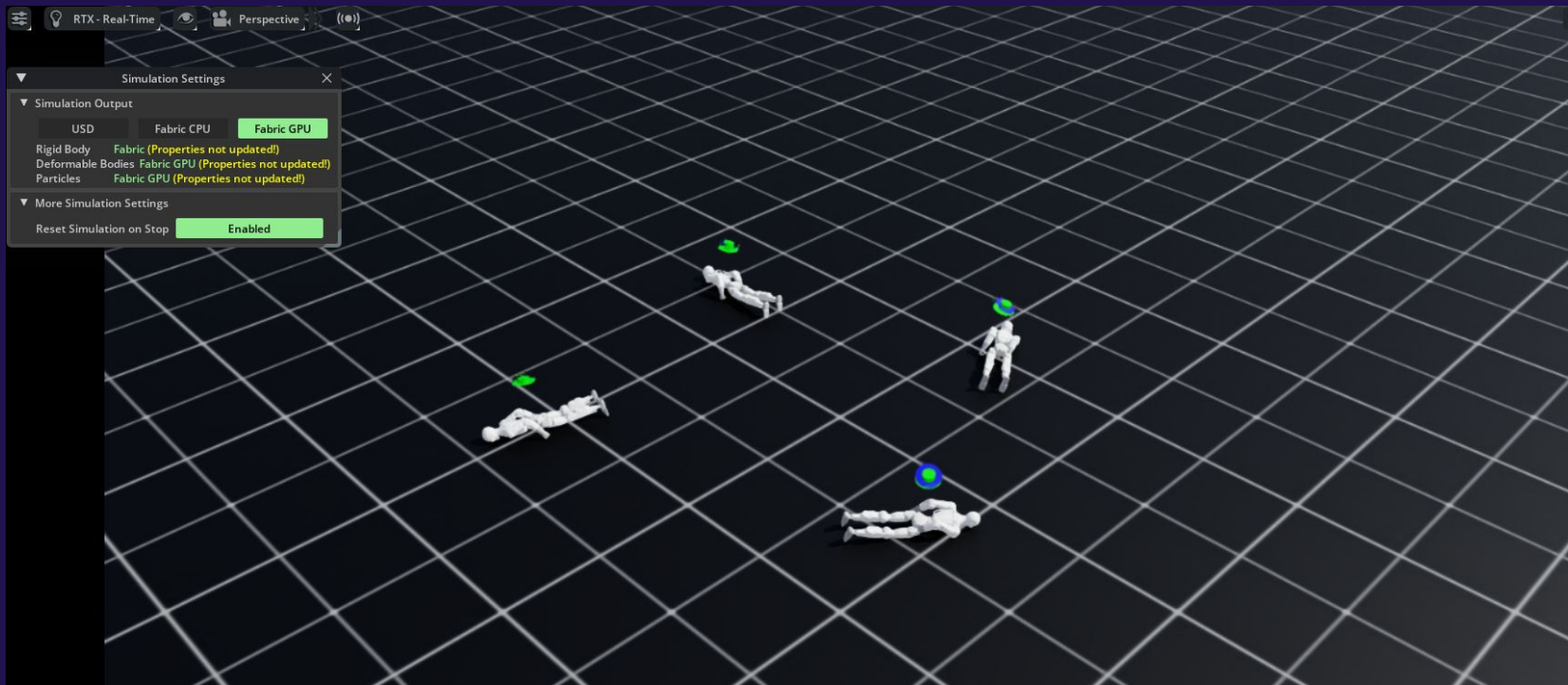
Training time: 3.79 seconds
PS C:\Users\Mario\IsaacLab>
```

Confirming *play.py* function

```
.\isaacclab.bat -p .\scripts\reinforcement_learning\rsl_rl\play.py --task  
Isaac-Velocity-Rough-R1-Play-v0 --num_envs 4
```

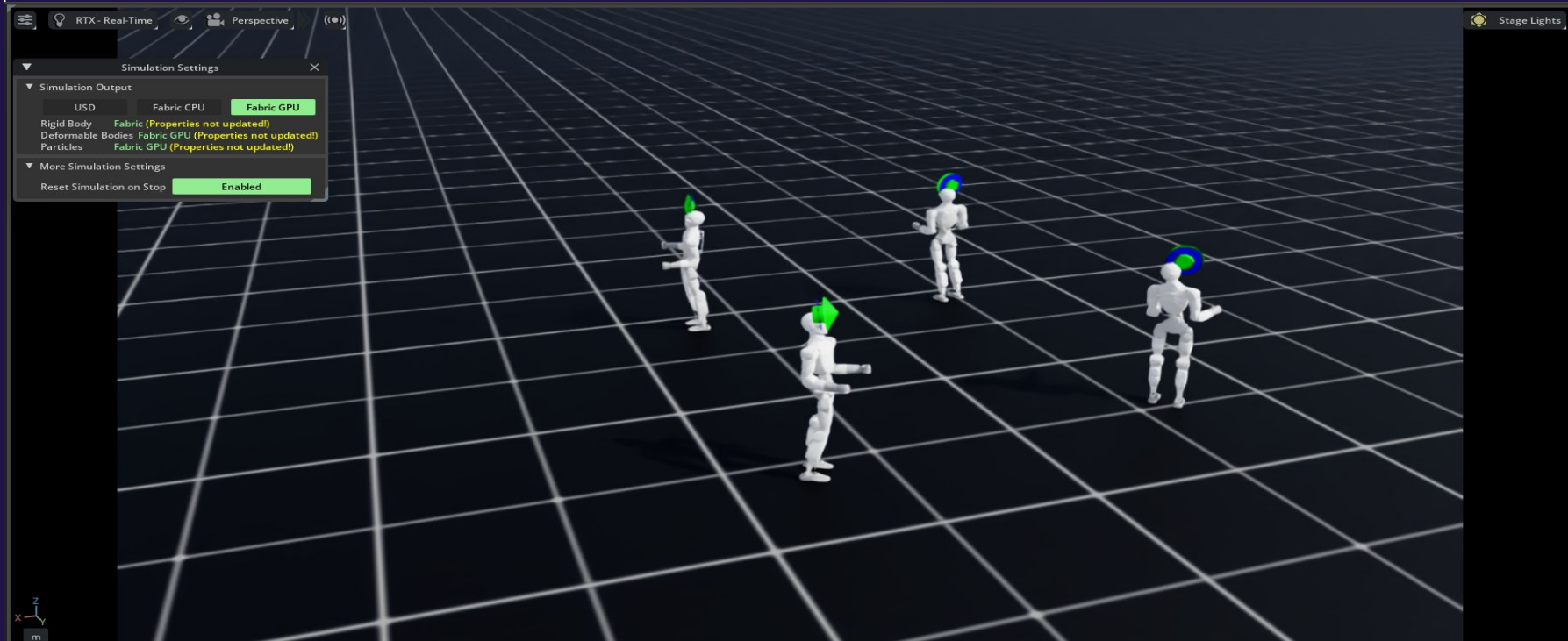


Confirming *play.py* function



Full Training

- `.\isaacclab.bat -p .\scripts\reinforcement_learning\rsl_rl\train.py --task Isaac-Velocity-Rough-R1-v0 --num_envs 1024 --headless`



Next Step

- Implementing rewards/penalties to make the R1 walk
 - Programming logic from the G1's `rough_env_cfg.py`